

to get a better short-circuit by using instead of a glass plate a silver plate, so that there can no longer be any difference of potential between the chromogelatine and the drops of silver nitrate. In these circumstances, the formation of Liesegang rings has not been observed; the formation of the periodic precipitate ceases, and instead of it a homogeneous diffuse precipitate appears (Fig. 4). This shows that the formation of a periodic precipitate in an isotropic colloidal medium is very similar to the phenomena of liquid-contact potentials. On eliminating this potential, periodic precipitation does not occur.

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Dipole Moments of some *Ortho*-Substituted Anilines

WE have measured the dipole moments of aniline, *N*-methyl aniline, *N,N*-dimethyl aniline, and corresponding toluidines and xylydines. As is seen from the accompanying table, alkylation affects the dipole moments of aniline and *p*-toluidine to a small extent, the difference in dipole moment between the primary and the tertiary amines being about 0.05 Debye. In the case of *ortho*-substituted amines, however, the substitution of two methyl groups at the nitrogen atom lowers the dipole moment by about 0.7 Debye. The dipole moments of these tertiary aniline homologues have almost the same value as that of aliphatic tertiary amines, namely, 0.8 Debye¹. This result is in good agreement with the result obtained by Thomson² from measurements of molecular refractivities. He calculated that the atomic refractivity of nitrogen in dimethyl 2,6-xylydine approaches very closely the value for nitrogen in a tertiary aliphatic amine. Furthermore, Remington³ has found from ultra-violet absorption spectra that the resonance is inhibited in *ortho*-substituted derivatives of dimethyl aniline.

	μ	Δ
Aniline	1.59	
Dimethyl aniline	1.55	0.04
<i>p</i> -Toluidine	1.35	
Dimethyl <i>p</i> -toluidine	1.30	0.05
<i>o</i> -Toluidine	1.60	
Dimethyl <i>o</i> -toluidine	0.9	0.7
2,4-Xylydine	1.34	
Dimethyl 2,4-xylydine	0.7	0.6
2,6-Xylydine	1.65	
Dimethyl 2,6-xylydine	1.0	0.7

The investigation of the dipole moments of the monomethyl aniline derivatives, which is being continued in this laboratory, does not show any pronounced difference between the *ortho*-substituted secondary amines and the corresponding non-substituted one. Full details of the measurements will be published elsewhere.

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¹ Ghosh, P. N., and Chatterjee, T. P., *Phys. Rev.*, **37**, 427 (1931).

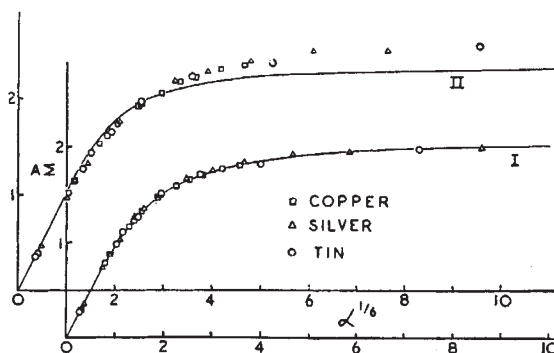
² Thomson, G., *J. Chem. Soc.*, 408 (1944).

³ Remington, W. R., *J. Amer. Chem. Soc.*, **67**, 1838 (1945).

Anomalous Skin Effect in Metals

AT sufficiently low temperatures and high frequencies, the mean free path of the electrons in a good conductor becomes greater than the classically predicted skin depth, and the classical skin-effect equations break down. Pippard¹ has studied the effect at liquid helium and liquid hydrogen temperatures, and Reuter and Sondheimer² have given the quantitative theory of the effect, and have interpreted Pippard's results. They found that for tin, mercury and aluminium, the experimental results implied reasonable values for the ratio σ/l of conductivity to electronic mean free path, but that for copper, silver and gold they implied values lower than the expected values by a factor of 5 or more. It was suggested that these low values were due to surface conditions differing from those in the bulk metal. In order to study this suggestion and to make a more detailed comparison with the theory than was possible on the basis of Pippard's measurements, a new investigation has been carried out in which attention has been paid to surface conditions, and the whole temperature range from 2° K. to 90° K. has been covered.

An electrically heated cryostat was built, capable when immersed in a bath of liquid helium or liquid hydrogen of maintaining any temperature between 2° K. and 90° K. for an indefinite period. The radio-frequency conductivity of the specimens was measured at about 1,200 Mc./s. by the method used by Pippard, and simultaneous measurements of the D.C. conductivity were made on similar specimens.



I, Experimental points fitted to $p = 0$ curve. II, Attempt to fit to $p = 1$ curve. Full lines theoretical

Measurements have so far been made on tin, cast in silica capillaries, and copper and silver, electrolytically polished. For all specimens, the results indicate that reflexion of electrons at the surface is completely diffuse, at all temperatures. The accompanying graphs show typical results on three specimens, scaled (I) to fit the theoretical (Reuter and Sondheimer) curve for completely diffuse reflexion ($p = 0$), and (II) in an attempt to fit the curve for specular reflexion ($p = 1$). It might have been expected that the 'reflexion coefficient' p would rise as the temperature was lowered and thermal disturbances at the surface became smaller: that this is not observed presumably indicates permanent surface irregularities of the order of magnitude of the electronic wave-length, that is, a few angstroms. The Reuter and Sondheimer curves relate two variables $A\Sigma$ and α ; here $A\Sigma$ is proportional to the observed Q of the specimen, the scaling factor depending on

the geometry of the resonator and the constant of the metal, and $\alpha = 3l^2/2\delta^2$, where l is the mean free path and δ the classical skin depth. Hence $\alpha^{1/3} \propto \sigma$, the d.c. conductivity; the constant of proportionality differs from metal to metal, and from it may be found the 'free-electron' values of σ/l and also of n_e/n_a , the effective number of conduction electrons per atom. The values thus deduced from the experiments on tin, copper and silver are given in the accompanying table (specimens Sn, Cu, Ag); the probable errors of these figures are about ± 5 per cent. Also given, for comparison, are the results deduced from Pippard's measurements (Sn (P), etc.), which have rather larger probable errors, and from measurements on two other specimens treated as described below (Cu A, Cu B).

Specimen	σ/l (ohm ⁻¹ cm. ⁻² $\times 10^{10}$)	n_e/n_a
Sn	9.5	1.1
Sn (P)	8.0	0.87
Ag	9.0	0.66
Ag (P)	1.7	0.052
Cu	17.0	1.17
Cu (P)	2.8	0.1
Cu A	2.9	0.1
Cu B	15.0	0.97

It will be seen that for tin the present results are of the same order as Pippard's result Sn (P), but that for copper and silver the present figures are much higher and much closer to the expected figures. Whereas Pippard's specimens were annealed but otherwise left unaltered after drawing down to the required diameter, specimens Cu and Ag were electrolytically polished to remove the surface layer. Two mechanisms have been suggested to account for the effect of the surface layer: (i) roughness, and (ii) increased resistivity owing to cold-work during drawing, not removed by subsequent annealing. To decide between these, specimens Cu A and Cu B were measured. Cu A, a specimen previously measured in the electro-polished condition, was treated with 'Brasso' metal polish. The surface still appeared to be very highly polished, but the effect on σ/l and n_e/n_a was drastic; they were reduced to about the same values as for Cu (P). Specimen Cu B, also previously electropolished, was dipped in nitric acid for a few moments, which produced a dull mat surface, showing under the microscope considerable roughness of the required order of magnitude, that is, a few thousand angstroms. It will be seen, however, that the effect on σ/l and n_e/n_a is small. It is therefore concluded that the predominant cause of high surface losses is a high-resistance surface layer. By analysing the form of the curve for Cu A, it was possible to deduce on this assumption that the surface layer extended for more than 2,000 angstroms below the surface, and had a residual resistivity about ten times that of the bulk metal.

Experiments are in progress on gold and the alkali metals, and also on metals other than Group I metals.

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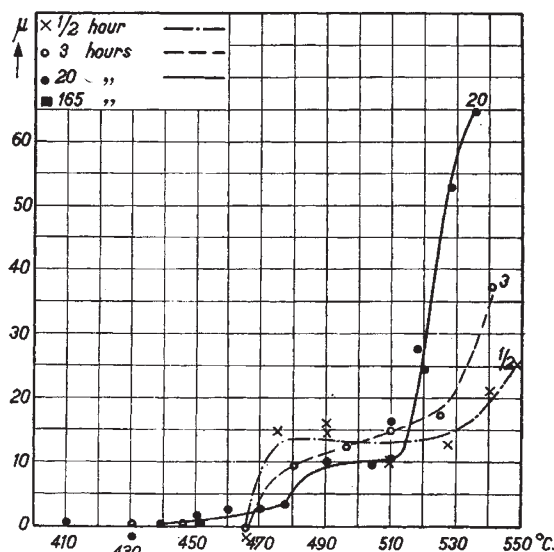
¹ Pippard, A. B., *Proc. Roy. Soc., A*, **191**, 385 (1947).

² Reuter, G. E. H., and Sondheimer, E. H., *Proc. Roy. Soc., A*, **195**, 336 (1948).

Rapid Oxidation of Metals and Alloys in the Presence of Molybdenum Trioxide

In studying heat-resistant iron-base and copper-base alloys at 750° C., we observed a pronounced increase of the rate of oxidation, when the atmosphere was contaminated by molybdenum trioxide vapour. In the meantime, this effect has been discovered independently by Leslie and Fontana¹, who observed above about 800° C. a rapid oxidation of steels with a high content of molybdenum. These authors ascribed the abnormal oxidation to a catalytic action of molybdenum trioxide, the thermal dissociation into molybdenum dioxide and nascent oxygen being the important factors in their opinion.

In connexion with the low melting point of molybdenum trioxide (790° C.) we wished to settle the question whether a liquid oxide phase hampers the development of a protective oxide film, thus causing the rapid oxidation. To this end we studied the temperature-dependence of the influence of molybdenum trioxide on the oxidation of different metals and alloys. Here we wish to restrict ourselves to a description of the influence of the trioxide on the oxidation of aluminium-copper, copper, silver and aluminium-silver. These materials are especially interesting because molybdenum trioxide markedly influences their rate of oxidation at such low temperatures (below 500° C.) that the thermal decomposition of molybdenum trioxide can be disregarded. At these temperatures even the vapour pressure of molybdenum trioxide is very small (at 500° C., $\sim 10^{-5}$ mm. mercury). Therefore the oxidation was carried out in such a way that the metallic specimens were at the same time in contact with air and with the trioxide powder. Weighing experiments, microscopic analysis and tensile tests of oxidized wires gave reasonable agreement as to the effective depth of penetration d , caused by the action of molybdenum trioxide. The values of d determined for 8/92 aluminium-copper in the accompanying graph refer to tensile tests and are defined as $d = r_0(1 - \sqrt{K_M/K_A})$, where r_0 is the original radius of the wire, while K_M and K_A are the forces required to rupture the wires



Penetration depth on oxidizing 8/92 aluminium-copper wires in the presence of molybdenum trioxide at different temperatures